Model Justification: For this project, two different types of classifier are being used, a distance-based one (SVC) and a tree-based one (Random Forest). The reasoning behind choosing a Random Forest classifier was to utilize the effects of feature selection and majority voting to have a more interpretable result and limit variance. This was especially important given the nature of the data, where the goal is to look at the impact features have on diabetic cases. By using Random Forest, the possibility of dominant features becoming the only source the model goes to for its predictions is limited.

The logic behind SVC was to compare what Random Forest does with something that is strictly distance-based, to see if concrete decision boundaries would separate the classes with different performance levels. The primary hope was to extract results that would either support the conclusions of random forest, and create a more robust analysis, and also identify which model was best at precision and recall (false negatives are extremely important).

K-Fold Validation: I choose a k of 5 because the training set had roughly 200,000 records, and I felt 40,000 records per fold would be sufficient for a multiclass classification. This decision also came as a result of tweaking with the under- and oversampling ratios, as well as considering other options to approach the issues regarding prediabetic classifications.

Parametric Tunings:

Random Forest-

n_estimators: the number of trees, more trees can reduce variance through averaging

max_depth: deeper trees are more complex, shallower trees prone to underfitting

min_samples_leaf: higher number means more records need to exist in the newly formed branches in order for the model to go deeper

max_features: amount of features considered per split, limits correlation

SVC-

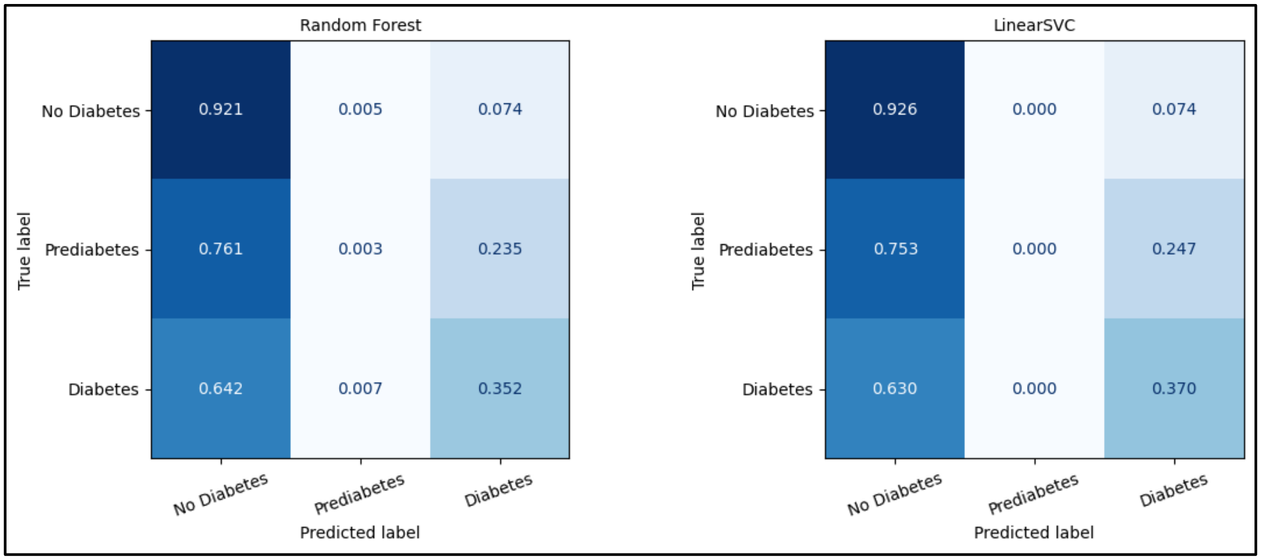
C: controls penalty for misclassifcations (margin for error)

Gamma: determines how much influence a single training point has

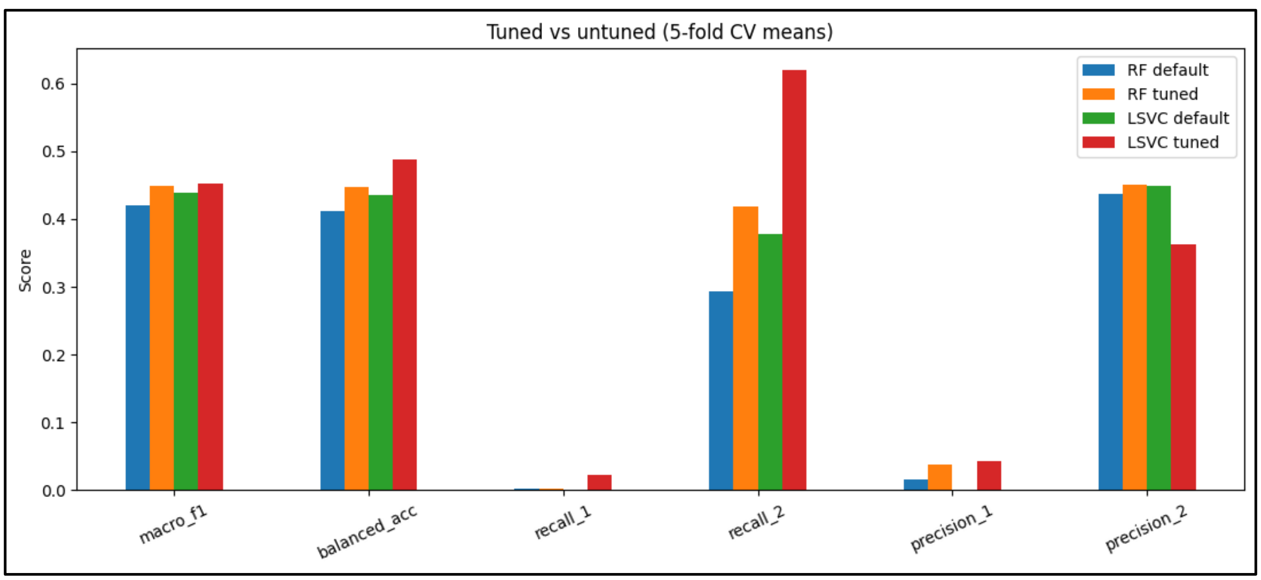
Kernel: function to map low-dimensional data into higher-dimensional space

Results:

Initial run of RF and SVC, default parameters



Recall 1, 2 and Precision 1, 2 correspond to classes 1 and 2



Potential Improvements:

I would like to continue testing over- and undersampling ratios, but I believe that the most likely course of action will be to drop the prediabetic class and turn this into a binary classification. Another option that could stem from this approach is to include the prediabetic cases as members of either class 0 or class 1, and look at the results of the models that way. While it may not be as concrete in its interpretability, if there are any major differences it could lead to discovery about the prediabetic class. Overall, I will most likely drop the prediabetic class and turn this project into one that strictly looks at diabetic vs non-diabetic cases.